

Diskrete Mathematik – Woche 2

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<https://github.com/tschai-yim/eth-resources>

September 27, 2026

Propositional Logic

Problem 1 ★

Evaluate the truth value of the formula $(A \wedge \neg A) \rightarrow B$ for any arbitrary truth assignment of B . Justify your answer algebraically using Lemma 2.1 and the useful equivalences.

Problem 2 ★

Determine whether the following expressions are syntactically valid formulas or statements about formulas. If an expression is invalid in both contexts, explain why.

1. $(A \rightarrow B) \equiv (\neg A \vee B)$
2. $(A \equiv B) \vee C$
3. $\models (A \leftrightarrow A)$
4. $A \leftrightarrow (B \leftrightarrow C)$

Problem 3 ★ ★

Assume that $F \models G$ holds true for two propositional formulas F and G . If G is known to be unsatisfiable ($\perp$), what must be true about the formula F ? Proof your statement.

Problem 4 ★

Prove or disprove the following claim: “For any formulas F and G , if the formula $F \rightarrow G$ is satisfiable, then $F \models G$.”

Problem 5 (skip for time) ★ ★

Using the equivalences from Lemma 2.1 (specifically the distributive and absorption laws), prove that $(A \vee B) \wedge (A \vee \neg B) \equiv A$.

Problem 6 (skip for time) ★ ★

Find a formula equivalent to $\neg(A \rightarrow (B \wedge C))$ that only uses the operators $\wedge$, $\vee$, and $\neg$, and where the $\neg$ operator only applies directly to the propositional symbols (literals).

Predicate Logic

Problem 7 ★ ★

The script explicitly defines the Universe U as a **non-empty** set. Consider the logical consequence: $\forall x P(x) \models \exists x P(x)$. Explain why this holds true under the definition provided, but would fail if U were allowed to be the empty set $\emptyset$.

Problem 8 ★ ★

Consider the formula $F = \forall x \exists y (y < x)$. Evaluate its truth value under the following interpretations:

1. $U = \mathbb{N}$ (natural numbers $0, 1, 2, \dots$) and $<$ is the standard “less than” predicate.
2. $U = \mathbb{Z}$ (integers) and $<$ is the standard “less than” predicate.

Problem 9 ★

Rewrite the following formulas using a new variable y so that the quantifier scopes do not overlap, preserving the exact same logical meaning.

1. $\exists x(x < 0) \wedge \exists x(x > 0)$
2. $\forall x(P(x) \rightarrow \exists x(Q(x)))$

Problem 10 ★ ★

Let U be the set of all animals. Let $C(x)$ mean “ x is a cat”, $M(x)$ mean “ x is a mammal”, and $T(x)$ mean “ x has a tail”. Translate the following statement into a single predicate logic formula: “All cats are mammals, but not all cats have tails.”

Problem 11 (skip for time) ★ ★ ★

Let $U = \mathbb{N}$. Let $E(x)$ be the predicate “ x is even”, and $P(x)$ be “ x is prime”. Using only the standard quantifiers ($\forall, \exists$) and equality ($=$), write a formula expressing: “There exists exactly one even prime number.”

Problem 12 ★ ★

Prove or disprove the following claims regarding logical consequence:

1. $\exists xP(x) \wedge \exists xQ(x) \models \exists x(P(x) \wedge Q(x))$
2. $\forall x(P(x) \vee Q(x)) \models \forall xP(x) \vee \forall xQ(x)$
3. $\forall xP(x) \wedge \forall xQ(x) \models \forall x(P(x) \wedge Q(x))$ (★ ★ ★)

Problem 13 ★ ★

Find a formula equivalent to $\neg(\forall x\exists y\forall z(P(x, y) \rightarrow Q(y, z)))$ where the negation operator $\neg$ only applies directly to the predicates P and Q . Show your step-by-step equivalence derivations.

Problem 14 (skip for time) ★

Let $U = \mathbb{Z}$ (integers) and $S(x, y)$ be the predicate $x + y = 0$. Determine the truth values of the statements $S_1 = \forall x\exists yS(x, y)$ and $S_2 = \exists y\forall xS(x, y)$.